

1. A 54-year-old woman with diabetes mellitus has had increasing shortness of breath and ankle swelling for 3 weeks. She has a nonproductive cough and discomfort when sleeping on fewer than four pillows. There is an S₃ with dullness to percussion and diminished breath sounds at the base of the right hemithorax. Which of the following is the most likely finding on examination of pleural fluid?
- ☐ A) ~~Erythrocyte count greater than 15,000/mm³~~
 - ☒ B) Glucose concentration approximately equal to serum glucose concentration
 - ☐ C) ~~Lactate dehydrogenase activity greater than or equal to 200 U/L~~
 - ☐ D) ~~Neutrophilic leukocyte count greater than 1000/mm³~~
 - ☐ E) Protein concentration greater than or equal to 3 g/dL

2. A 52-year-old woman comes to the emergency department because of increasingly severe right knee pain since she tripped and fell outside her home 1 week ago. Initially, mild pain and swelling developed in the joint, but she was able to walk. The pain has worsened during the past 4 days, and she is now unable to bear weight on her right side. She has a 5-year history of type 2 diabetes mellitus treated with metformin. Her temperature is 37°C (98.6°F). Examination of the right knee shows warmth and a small effusion. Passive and active range of motion is limited by pain. The remainder of the examination shows no abnormalities. Her leukocyte count is 10,000/mm³, and serum erythrocyte sedimentation rate is 40 mm/h. An x-ray of the knee shows degenerative joint disease and osteopenia but no fracture. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Serum rheumatoid factor assay
 - ☐ B) Triple-phase technetium 99m bone scan
 - ☐ C) MRI of the knee
 - ☒ D) Aspiration of the knee joint
 - ☐ E) Bone biopsy

3. A 27-year-old woman comes to the physician because of a 2-month history of progressive fatigue. She has had excessive menstrual bleeding over the past year; she is otherwise healthy. Physical examination shows pale conjunctivae. The remainder of the examination, including pelvic examination, shows no abnormalities. Laboratory studies show:

Hemoglobin	10.2 g/dL
Leukocyte count	7600/mm ³
Segmented neutrophils	61%
Eosinophils	3%
Lymphocytes	31%
Monocytes	5%
Platelet count	410,000/mm ³

Which of the following additional laboratory abnormalities is most likely in this patient?

- ☒ A) Decreased mean red cell volume
- ☐ B) Decreased red cell distribution width
- ☐ C) Decreased serum total iron-binding capacity
- ☐ D) Increased reticulocyte count
- ☐ E) Increased serum ferritin concentration

4. A 62-year-old woman is scheduled to undergo lumpectomy and sentinel lymph node biopsy for breast cancer. She feels well. She has hypertension treated with an angiotensin-converting enzyme (ACE) inhibitor. She underwent coronary artery bypass grafting (CABG) 3 years ago and sustained a myocardial infarction (MI) 5 months ago. She has smoked one pack of cigarettes daily for 40 years. Her pulse is 65/min, respirations are 16/min, and blood pressure is 130/80 mm Hg. The lungs are clear to auscultation. Heart sounds are normal. An ECG shows five premature ventricular contractions per minute; there are no ischemic changes. Which of the following is the strongest predisposing factor for perioperative MI in this patient?
- ☐ A) Age
 - ☐ B) ECG findings
 - ☐ C) History of CABG
 - ☐ D) History of hypertension
 - ☐ E) History of smoking
 - ☒ F) MI within the past 6 months

5. A 37-year-old man comes to the emergency department because of a 2-day history of abdominal cramps and diarrhea. He has vomited blood twice since the onset of symptoms. He has been taking ibuprofen for pain since he sustained a knee injury in a basketball game 1 month ago. He appears pale. His temperature is 36.8°C (98.2°F), pulse is 130/min, respirations are 22/min, and blood pressure is 110/70 mm Hg. The lungs are clear to auscultation. The abdomen is soft and nontender. The stool is dark and tarry, and test for occult blood is positive. Two liters of 0.9% saline are administered. Laboratory studies show:

Hematocrit	18%
Leukocyte count	6200/mm ³
Platelet count	250,000/mm ³
Prothrombin time	12 sec
	(INR=1)
Partial thromboplastin time	35 sec

The most appropriate next step in management is administration of which of the following?

- ☐ A) Additional 2 L of 0.9% saline
- ☐ B) Fresh frozen plasma
- ☐ C) H₂-receptor blocking agent
- ☐ D) Octreotide
- ☒ E) Packed red blood cells

6. A previously healthy 32-year-old woman comes to the physician 1 day after a rash developed on her face, neck, and hands. Prior to onset of the symptoms, she was weeding and fertilizing her backyard while wearing a sleeveless shirt and shorts. She used a sunscreen but did not wear a hat or use insect repellent. She reports that she also received several scratches on her hands from the rose bushes. Examination shows bright red papules, vesicles, and bullae, some in linear pattern, on her forearms, neck, and face. There are oozing vesicles over the wrists. Which of the following is most likely to have prevented this rash?
- ☐ A) Avoidance of contact with fertilizers
- ☐ B) Avoidance of contact with rose thorns
- ☐ C) Avoidance of contact with weeds
- ☒ D) Avoidance of sunscreen
- ☐ E) Wearing insect repellent

Contact with weeds - hiram. Contact dermatitis.

7. An 18-year-old woman has had fever, sore throat, nausea, and anorexia for 5 days. Her temperature is 39°C (102.2°F). She has exudative pharyngitis with enlarged tonsils and prominent anterior and posterior cervical lymphadenopathy. A grade 1/6, systolic ejection murmur is heard. The spleen tip is palpated 1 cm below the left costal margin. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Antibody titer to toxoplasmosis
 - ☐ B) Antistreptolysin O titer
 - ☐ C) Blood culture
 - ☒ D) Monospot test
 - ☐ E) Lymph node biopsy

8. A 57-year-old man comes to the physician because of a 4-day history of progressive shortness of breath and cough productive of yellow sputum. He has chronic obstructive pulmonary disease treated with an ipratropium inhaler, but he states that the inhaler has not relieved his shortness of breath over the past month. He smoked two packs of cigarettes daily for 40 years but quit 6 months ago. He has worked as a volunteer in a homeless shelter for the past 10 years. His temperature is 37°C (98.6°F), pulse is 110/min, respirations are 22/min, and blood pressure is 130/80 mm Hg. Breath sounds are decreased. Scattered rhonchi are heard. Pulse oximetry on room air shows an oxygen saturation of 90%. An x-ray of the chest shows hyperinflation of the lungs. Which of the following is the most likely diagnosis?
- ☐ A) Asthma
 - ☒ B) Bronchitis
 - ☐ C) Bronchogenic carcinoma
 - ☐ D) Community-acquired pneumonia
 - ☐ E) Tuberculosis

9. A 52-year-old woman comes to the physician because of a 7-day history of urinary frequency. She says that recently she has been thirsty during the day and has been drinking more fluids than usual but limits fluids before bed and still urinates four to six times at night. She has an 8-year history of hypertension treated with hydrochlorothiazide (25 mg daily), which she takes in the morning. Menopause occurred 3 years ago. She does not take any other medications or vitamin supplements. She has been sexually active with one male partner for 20 years and no longer uses contraception. Her blood pressure is 128/80 mm Hg. Examination shows no abnormalities. Serum studies show:

Na ⁺	148 mEq/L
Cl ⁻	110 mEq/L
K ⁺	3.9 mEq/L
HCO ₃ ⁻	26 mEq/L
Urea nitrogen	20 mg/dL
Creatinine	1.0 mg/dL

Results of a water deprivation test show:

Time (h)	Urine Osmolality (mOsmol/kg)	Serum Osmolality (mOsmol/kg)
0	200	285
1	250	288
2	300	290
3	380	295
4	400	298
5	400	300

One hour after administration of desmopressin, her urine osmolality is 1000 mOsmol/kg. Which of the following is the most likely cause of this patient's polyuria?

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- ☐ A) Hydrochlorothiazide-induced nephrogenic diabetes insipidus
- ☒ B) Idiopathic central diabetes insipidus
- ☐ C) Idiopathic nephrogenic diabetes insipidus
- ☐ D) Primary polydipsia
- ☐ E) Salt-losing nephropathy

10. An overweight 65-year-old man smokes one pack of cigarettes daily. His brother died of a cerebral infarction, and both of his parents died while in their 40s. His blood pressure is 150/102 mm Hg, and he has grade 2 hypertensive retinopathy and a point of maximal cardiac impulse 2 cm lateral to the midclavicular line. As part of the initial assessment, which of the following laboratory tests is most appropriate?

- ☐ A) Measurement of plasma renin activity
- ☐ B) Measurement of serum aldosterone concentration
- ☒ C) Measurement of serum creatinine concentration
- ☐ D) Measurement of urine metanephrine concentration
- ☐ E) Urinary sodium/creatinine ratio

11. Two weeks after admission to the hospital for treatment of an unexplained fever, a 57-year-old man develops acute renal failure. Current medications include acetazolamide, lisinopril, heparin, amikacin, and naproxen. Examination shows 2+ pitting edema of the lower extremities below the midcalf level. Urinalysis shows a few WBCs, epithelial cells, and numerous muddy brown casts. Which of the following medications is the most likely cause of his acute renal failure?

- ☐ A) Acetazolamide
- ☒ B) Amikacin
- ☐ C) Heparin
- ☐ D) Lisinopril
- ☐ E) Naproxen

12. A 37-year-old man comes to the physician because of an 8-week history of diarrhea; he has had 8 to 10 watery bowel movements daily during this period and has also noted hair loss. Four weeks ago, he noticed a rash on his elbows and knees. He has a 6-year history of regional enteritis treated on several occasions with 2- to 3-month courses of oral corticosteroids. Examination shows patchy alopecia on the scalp and hyperkeratotic lesions on the elbows and knees. This patient's condition is most likely the result of a deficiency of which of the following?

- ☐ A) Cobalt
- ☐ B) Copper
- ☐ C) Manganese
- ☐ D) Selenium
- ☒ E) Zinc

13. A 67-year-old man comes to the physician because of a 3-day history of fever, cough productive of dark sputum, pleuritic chest pain, and shortness of breath. One week prior to the onset of symptoms, he had a runny nose, nasal congestion, and mild sore throat. His temperature now is 38.4°C (101.2°F), pulse is 92/min, respirations are 18/min, and blood pressure is 146/86 mm Hg. Bronchial breath sounds are heard in the left lower lung field; there is dullness to percussion. An x-ray of the chest shows a left lower lobe infiltrate and a pleural effusion. Thoracentesis yields yellow fluid. Laboratory studies show:

Serum	
Glucose	105 mg/dL
Protein	7.2 g/dL
Lactate dehydrogenase (LDH)	55 U/L
Pleural fluid	
Erythrocyte count	1025/mm ³
Leukocyte count	3500/mm ³
Segmented neutrophils	90%
Glucose	80 mg/dL
Protein	6.5 g/dL
Triglycerides	30 mg/dL
Amylase	30 U/L
LDH	95 U/L
pH	6.9

A Gram stain of the fluid shows segmented neutrophils and gram-positive diplococci. Culture of the fluid is pending. Which of the following is the most likely diagnosis?

- ☐ A) Chylothorax
- ☐ B) Congestive heart failure
- ☒ C) Empyema
- ☐ D) Tuberculous pleuritis
- ☐ E) Uncomplicated parapneumonic effusion

14. A previously healthy 27-year-old man comes to the physician because of fatigue and loss of appetite for 2 weeks. He does not smoke and drinks a small amount of alcohol on social occasions. His mother has hemochromatosis. Examination shows scleral icterus. The liver edge is smooth and tender and palpated 3 cm below the right costal margin. Laboratory studies show:

Hemoglobin	14.3 g/dL
Serum	
Albumin	3.7 g/dL
Total bilirubin	4.4 mg/dL
Alkaline phosphatase	176 U/L
AST	149 U/L
ALT	215 U/L
Hepatitis B surface antigen	negative
Anti-hepatitis B core antigen	positive
Anti-hepatitis B surface antigen	negative

Which of the following is most likely to have prevented this patient's jaundice?

- ☐ A) Abstinence from alcohol
- ☐ B) Good hand washing
- ☐ C) Ophthalmic screening for the presence of Kayser-Fleischer rings
- ☐ D) Transferrin saturation testing
- ☒ E) Administration of hepatitis B vaccine

15. A 20-year-old man comes to the physician because his friends tease him about having large breasts. His breasts have increased in size since he was approximately 12 years old. He has no history of serious illness and takes no medications. He says he needs to shave only once or twice weekly, does not have much sexual interest in men or women, and rarely has erections. He is not sexually active. He is in the third year of an undergraduate honors program in chemical engineering. He is 186 cm (6 ft 1 in) tall and weighs 66 kg (145 lb); BMI is 19 kg/m². Examination shows sparse facial hair and long upper and lower extremities. There is marked bilateral gynecomastia. Pubic hair development is Tanner stage 3. The testes are small. Serum studies show a follicle-stimulating hormone (FSH) concentration of 50 mIU/mL, luteinizing hormone (LH) concentration of 40 mIU/mL, and testosterone concentration of 6 nmol/L (N=10–35). Which of the following is most likely to confirm the diagnosis?
- ☐ A) DNA analysis for trinucleotide repeats
 - ☐ B) Growth hormone stimulation test
 - ☐ C) Growth hormone suppression test
 - ☒ D) Karyotype analysis
 - ☐ E) Measurement of serum concentrations of FSH, LH, and testosterone after intravenous administration of gonadotropin-releasing hormone
 - ☐ F) Measurement of serum estradiol concentration
 - ☐ G) Measurement of serum prolactin concentration
 - ☐ H) Measurement of serum thyroid-stimulating hormone concentration
 - ☐ I) MRI of the hypothalamus and pituitary gland

16. A 90-year-old woman comes to the physician because of a 1-month history of **generalized itching** and a decreased appetite resulting in a 4.5-kg (10-lb) weight loss. She has not had nausea or vomiting. She has a 30-year history of hypertension treated with hydrochlorothiazide and atenolol. She has no difficulty with normal activities of daily living and sleeps 8 hours nightly. She appears frail and is not in distress. She is 168 cm (5 ft 6 in) tall and weighs 40 kg (88 lb); BMI is 14 kg/m². Her pulse is 70/min and regular, and blood pressure is 160/80 mm Hg. Examination shows **excoriations over the upper and lower extremities**. Mental status examination shows a normal affect. Cognition is intact. Laboratory studies show:

Hemoglobin	10 g/dL
Leukocyte count	4000/mm ³
Serum	
Na ⁺	138 mEq/L
K ⁺	5.9 mEq/L
Cl ⁻	110 mEq/L
HCO ₃ ⁻	20 mEq/L
Ca ²⁺	10.2
	mg/dL
Urea nitrogen	45 mg/dL
Creatinine	3 mg/dL

Which of the following is the most likely cause of this patient's anorexia?

- ☐ A) Adverse effect of atenolol
- ☐ B) Hyperkalemia
- ☐ C) Hyperthyroidism
- ☐ D) Major depressive disorder
- ☒ E) Renal failure

17. A 25-year-old farmworker is brought to the emergency department because of the sudden onset of headache, dizziness, nausea, and vomiting while picking peaches. His pulse is 52/min, and blood pressure is 90/60 mm Hg. He has generalized weakness, diaphoresis, and pinpoint pupils. Which of the following is the most likely mechanism of this condition?
- ☐ A) Decreased release of acetylcholine from presynaptic vesicles
 - ☐ B) Decreased stimulation of postsynaptic receptors by acetylcholine
 - ☒ C) Decreased synaptic activity of cholinesterase
 - ☐ D) Increased metabolic breakdown of norepinephrine in the synapse
 - ☐ E) Release of norepinephrine from sympathetic nerve endings

The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with chest pain, select the most appropriate next step in management.

- | | |
|--|---|
| ☐ A) Administration of a β -adrenergic blocking agent | ☐ H) Sublingual administration of nitroglycerin |
| ☐ B) CT scan of the chest | ☐ I) Ultrasonography of the abdomen |
| ☐ C) Hepatobiliary scan | ☐ J) Upper endoscopy |
| ☒ D) Oral administration of an H_2 -receptor blocking agent | ☐ K) Upper gastrointestinal series |
| ☐ E) Oral administration of a nonsteroidal anti-inflammatory agent | ☐ L) Ventilation-perfusion lung scans |
| ☐ F) Outpatient exercise stress test | ☐ M) X-ray of the chest |
| ☐ G) Reassurance only | |

18. A previously healthy 32-year-old woman comes to the emergency department because of intermittent burning chest pain in her midsternum during the past 2 weeks. The pain is worse at night and often awakens her from sleep; her symptoms are relieved after sitting upright or walking around for 10 minutes. During the episodes, she has nausea but no shortness of breath or palpitations. She takes an oral contraceptive and has smoked one pack of cigarettes daily for 10 years. Her father had a myocardial infarction at the age of 60 years, and her paternal uncle had a myocardial infarction at the age of 58 years. Her temperature is 36.8°C (98.2°F), pulse is 80/min, respirations are 16/min, and blood pressure is 120/70 mm Hg. The lungs are clear to auscultation. A grade 2/6 holosystolic murmur is heard best at the apex and radiates to the axilla. Abdominal examination shows mild tenderness in the right upper quadrant with no masses or organomegaly. Bowel sounds are normal.

For each patient with chest pain, select the most appropriate next step in management.

- | | |
|--|---|
| ☐ A) Administration of a β -adrenergic blocking agent | ☐ H) Sublingual administration of nitroglycerin |
| ☐ B) CT scan of the chest | ☐ I) Ultrasonography of the abdomen |
| ☐ C) Hepatobiliary scan | ☐ J) Upper endoscopy |
| ☐ D) Oral administration of an H_2 -receptor blocking agent | ☐ K) Upper gastrointestinal series |
| ☐ E) Oral administration of a nonsteroidal anti-inflammatory agent | ☐ L) Ventilation-perfusion lung scans |
| ☐ F) Outpatient exercise stress test | ☐ M) X-ray of the chest |
| ☒ G) Reassurance only | |

19. A 15-year-old girl comes to the physician because of a 2-week history of intermittent, fleeting pain under her left breast. She has had no cough, nausea, or shortness of breath. Her uncle had a myocardial infarction at the age of 40 years, and her father had a myocardial infarction at the age of 38 years. Her temperature is 36.3°C (97.4°F), pulse is 96/min, respirations are 25/min, and blood pressure is 120/74 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a normal S_1 and S_2 and a **midsystolic click**. There is no chest wall tenderness. Pulse oximetry on room air shows an oxygen saturation of 98%. An ECG shows normal findings.

20. A 37-year-old woman is brought to the emergency department by paramedics 45 minutes after the sudden onset of nonradiating substernal chest pain that awakened her from sleep. She has had three similar episodes over the past year; each episode lasted about 20 minutes and resolved spontaneously. She has not had nausea, vomiting, diaphoresis, palpitations, heartburn, or abdominal pain. En route to the emergency department, an ECG showed sinus bradycardia with a normal axis, normal PR interval, and a QRS interval of 0.16 ms. There were 2-mm ST-segment elevations in leads II, III, and aVF with no Q waves. She has a 1-year history of migraines. Over the past 6 months, she has had a decrease in the frequency of migraines with atenolol therapy. She does not smoke cigarettes, drink alcohol, or use illicit drugs. She appears anxious and uncomfortable. Her pulse is 55/min and regular, respirations are 14/min, and blood pressure is 130/80 mm Hg. The remainder of the examination shows no abnormalities. **An ECG now shows no abnormalities.** Which of the following is the most likely cause of this patient's chest pain?
- ☐ A) Atherosclerotic coronary artery disease
 - ☐ B) Cholecystitis
 - ☐ C) Esophageal-spasm
 - ☐ D) Panic disorder
 - ☐ E) Pericarditis
 - ☐ F) Sick-sinus-syndrome
 - ☒ G) Variant angina pectoris

21. A 78-year-old woman has had fever, headache, malaise, and right flank pain for 4 days; the pain is exacerbated by coughing. Examination of the right flank shows the closely grouped cluster of eroded erythematous vesicles shown. Which of the following is the most likely causal organism?

- ☐ A) Cytomegalovirus
- ☐ B) Herpes simplex virus 1
- ☐ C) *Mycoplasma pneumoniae*
- ☐ D) *Streptococcus pneumoniae*
- ☒ E) Varicella-zoster virus



22. A previously healthy 62-year-old woman comes to the physician because of a 3-month history of shortness of breath and chest pain with exertion. Her pulse is 80/min and regular, respirations are 14/min, and blood pressure is 120/74 mm Hg. Cardiac examination shows a **slow-rising**, sustained carotid pulse. The apical beat is sustained and inferiorly and laterally displaced on palpation of the chest. A grade 4/6, harsh systolic ejection murmur is heard; it peaks in late systole. Which of the following is the most likely diagnosis?

- ☐ A) Atrial septal defect
- ☒ B) Bicuspid aortic valve
- ☐ C) Coarctation of the aorta
- ☐ D) Dextrocardia
- ☐ E) Rheumatic valvular disease
- ☐ F) Ventricular septal defect

23. A 47-year-old woman comes for a routine examination. She has a 6-year history of type 2 diabetes mellitus treated with glipizide. Her pulse is 72/min, and blood pressure is 138/88 mm Hg. The remainder of the examination, including skin and neurologic examinations, shows no abnormalities. Laboratory studies show:

Hemoglobin A _{1c}	5.8%
Serum	
Glucose, fasting	132 mg/dL
Creatinine	1 mg/dL
Cholesterol, total	170 mg/dL
HDL-cholesterol	45 mg/dL
LDL-cholesterol	95 mg/dL
Triglycerides	150 mg/dL

Urinalysis is within normal limits. The addition of which of the following medications is most appropriate to further decrease this patient's risk for diabetic complications?

- ☐ A) Cholestyramine
- ☐ B) Enalapril
- ☐ C) Insulin
- ☐ D) Metformin
- ☐ E) Simvastatin
- ☒ F) No additional medications indicated

Enalapril - hiram. New guidelines state <140, but they changed recently, so goal is <130/80

24. A 27-year-old man who is HIV positive comes to the physician for a follow-up examination. He feels well. Medications include antiretroviral agents and a double-strength tablet of trimethoprim-sulfamethoxazole daily. Laboratory studies 1 week ago showed a CD4+ T-lymphocyte count of $110/\text{mm}^3$ (Normal ≥ 500), a positive *Toxoplasma* IgG antibody assay, and a positive hepatitis B surface antibody assay. Which of the following is the most appropriate next step to assess this patient's risk for illness related to opportunistic infection?
- ☒ A) PPD skin test
 - ☐ B) Sputum cytology
 - ☐ C) Urine cytomegalovirus antigen assay
 - ☐ D) Stool culture for *Mycobacterium avium-intracellulare* complex
 - ☐ E) X-ray of the chest

25. A 62-year-old woman comes to the physician because of a 3-month history of mildly increased shortness of breath. She has a 3-year history of chronic obstructive pulmonary disease. She smoked one pack of cigarettes daily for 40 years but quit 3 years ago when she was diagnosed. She has not required pharmacotherapy. Her temperature is 36.7°C (98°F), pulse is 86/min, respirations are 16/min, and blood pressure is 138/76 mm Hg. Pulmonary examination shows mildly decreased air movement; an occasional wheeze is heard. Arterial blood gas analysis on room air shows:

pH	7.41
Pco ₂	40 mm Hg
Po ₂	74 mm Hg

Which of the following is the most appropriate next step in management?

- ☐ A) Pulmonary rehabilitation
- ☐ B) Home oxygen therapy
- ☒ C) Ipratropium therapy
- ☐ D) Prednisone therapy
- ☐ E) Theophylline therapy

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☒ Mark

National Board of Medical Examiners
Medicine Self-Assessment

Time Remaining:
1 hr 8 min 49 sec

26. A 37-year-old woman comes to the physician because of a 6-week history of puffy eyes and swelling of her legs. During the past 6 months, she has had heavy bleeding with menses and increasing pain in her arms and legs. Menses occur at regular 28-day intervals. She has no history of serious illness. Current medications include calcium carbonate and a multivitamin. She has had a 5.4-kg (12-lb) weight gain during the past 3 months. She is 168 cm (5 ft 6 in) tall and weighs 70 kg (155 lb); BMI is 25 kg/m². Her pulse is 55/min, and blood pressure is 100/70 mm Hg. Examination shows periorbital edema. Heart sounds are distant. Examination of the upper and lower extremities shows mild tenderness to palpation. There is 2+ edema to the midcalves. Pelvic examination shows no abnormalities. Serum studies show:

Na ⁺	130
	mEq/L
K ⁺	3.8
	mEq/L
Urea nitrogen	10
	mg/dL
Creatinine	1 mg/dL
Cholesterol	300
	mg/dL
Creatine kinase	130 U/L

Which of the following is the most appropriate next step in diagnosis?

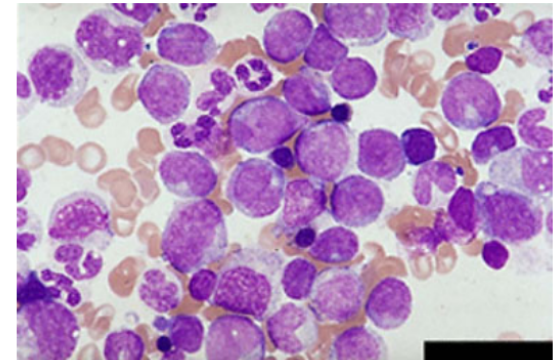
- ☐ A) HIV antibody testing
- ☐ B) Measurement of serum CA 125 concentration
- ☐ C) Measurement of serum follicle-stimulating hormone concentration
- ☐ D) Measurement of serum glucose concentration
- ☒ E) Measurement of serum thyroid-stimulating hormone concentration
- ☐ F) Muscle biopsy

27. A 47-year-old woman comes to the physician because of a nonproductive cough for 10 weeks. Six months ago, she had deep venous thrombosis treated with warfarin. She has never smoked. Her temperature is 37°C (98.6°F), pulse is 88/min, and blood pressure is 120/78 mm Hg. Examination shows no abnormalities. An x-ray of the chest shows a 3-cm peripheral lesion. Which of the following is the most likely diagnosis?

- ☒ A) Adenocarcinoma of the lung
- ☐ B) Bronchiolitis obliterans with organizing pneumonia
- ☐ C) Chronic idiopathic pulmonary fibrosis
- ☐ D) Pneumoconiosis
- ☐ E) Sarcoidosis
- ☐ F) ~~Small-cell lung cancer~~
- ☐ G) ~~Squamous-cell lung carcinoma~~
- ☐ H) Tuberculosis

28. A 37-year-old man comes to the physician because of a 6.8-kg (15-lb) weight loss during the past 3 months. He says he has a good appetite but cannot finish meals because he feels full prematurely. He has no history of serious illness and takes no medications. He is 165 cm (5 ft 5 in) tall and weighs 64 kg (142 lb); BMI is 24 kg/m². Vital signs are within normal limits. The spleen tip is palpated 3 cm below the left costal margin. Laboratory studies show:

Hematocrit	45%
Leukocyte count	23,600/mm ³
Segmented neutrophils	50%
Bands	15%
Metamyelocytes	5%
Eosinophils	2%
Basophils	3%
Lymphocytes	15%
Myelocytes	10%
Platelet count	500,000/mm ³



A blood smear is shown. Which of the following is the most likely cause of these findings?

- ☐ A) Chronic lymphocytic leukemia
- ☒ B) Chronic myelogenous leukemia
- ☐ C) Hairy cell leukemia
- ☐ D) Leukemoid reaction
- ☐ E) Splenic vein thrombosis

29. A 52-year-old obese woman is brought to the emergency department because of severe shortness of breath for 1 hour. She was healthy until 5 days ago, when she had moderate shortness of breath with exertion. She has no history of similar symptoms or asthma, pulmonary, or cardiac problems. She had an upper respiratory tract infection 2 weeks ago. She has never smoked. Her only medication is fluoxetine for major depressive disorder. On arrival, she refuses to recline on the gurney because she cannot breathe. Her temperature is 37.3°C (99.2°F), pulse is 100/min and regular, respirations are 24/min, and blood pressure is 142/88 mm Hg. Examination shows jugular venous distention. Bilateral crackles are heard one third of the way up the posterior lung fields. Examination shows a soft abdomen. There is 1+ pitting edema of the lower extremities. Two liters per minute of oxygen via nasal cannula are administered. An x-ray of the chest is ordered. Which of the following is the most appropriate next step in management?

- ☐ A) Inhaled albuterol
- ☐ B) Intravenous captopril
- ☐ C) Intravenous digoxin
- ☐ D) Intravenous dobutamine
- ☒ E) Intravenous furosemide
- ☐ F) Intravenous metoprolol
- ☐ G) Intravenous 0.9% saline

30. A 22-year-old man remains in the hospital 2 days after admission for a closed head injury sustained in a motor vehicle collision. A CT scan of the head on admission showed no abnormalities. He was initially confused, but he has become well oriented with normal vital signs during the past 12 hours. He is currently receiving 5% dextrose in 0.45% saline (50 mL/h). He has started to eat solids during the past 6 hours. His average urine output has been 50 mL/h during the past 24 hours. He is alert. Vital signs are within normal limits. Examination shows an ecchymosis over the frontal region. The remainder of the examination, including neurologic examination, shows no abnormalities. His serum sodium concentration is 120 mEq/L, and urine osmolality is 340 mOsm/kg. Which of the following is the most appropriate next step in management?
- ☒ A) Fluid restriction
 - ☐ B) CT scan of the head
 - ☐ C) ADH (vasopressin) therapy
 - ☐ D) Bolus of 0.9% saline
 - ☐ E) Bolus of 3% saline

31. A 37-year-old woman comes to the physician because of a 1-year history of severe heartburn; use of over-the-counter ranitidine and antacids has not resolved her symptoms. She has had intermittent diarrhea and a 2.3-kg (5-lb) weight loss during this period. She also reports that her fingers lose color when she takes food from the freezer. Her vital signs are within normal limits. Examination shows painful edema of several fingertips. Abdominal and rectal examinations show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Amyloidosis
- ☐ B) Carcinoid syndrome
- ☐ C) Systemic lupus erythematosus
- ☒ D) Systemic sclerosis (scleroderma)
- ☐ E) Zollinger-Ellison syndrome

32. A 42-year-old woman with newly diagnosed polycystic kidney disease (PKD) comes to the physician for a follow-up examination. She asks if any of her three children will inherit PKD. Her parents and three siblings do not have this disorder nor does the children's father. The risk for PKD in her children is closest to which of the following?

- ☒ A) 50% for each child, since PKD is autosomal dominant
- ☐ B) 25% for each child, since PKD is controlled by several genes
- ☐ C) 100% for each male child, since PKD is X-linked
- ☐ D) No risk, since the patient represents a new mutation
- ☐ E) No risk, since PKD is not a genetic disorder

33. A 47-year-old man comes to the physician because of a 2-week history of fever and cough productive of brown, foul-smelling sputum. He has had a 2.7-kg (6-lb) weight loss during this period. He has long-standing alcoholism. He takes no medications. His temperature is 38°C (100.4°F). Examination shows foul-smelling breath, poor dentition, and numerous dental caries. Breath sounds are decreased over the right lower lung field. An x-ray of the chest shows a 3 x 5-cm, round lesion in the right lower lung with an air-fluid level. Which of the following is the most appropriate pharmacotherapy?
- ☐ A) Azithromycin
 - ☒ B) Clindamycin
 - ☐ C) Fluconazole
 - ☐ D) Gentamicin
 - ☐ E) Trimethoprim-sulfamethoxazole

34. A 57-year-old woman is brought to the emergency department 4 hours after the onset of confusion and drowsiness. She has a 2-day history of nausea and vomiting and generalized weakness. She received the diagnosis of breast carcinoma metastatic to bone 1 year ago and has received pamidronate and tamoxifen since then. On arrival, she appears lethargic and is not oriented to person, place, or time. Her pulse is 78/min, respirations are 18/min, and blood pressure is 140/92 mm Hg. Examination shows sunken eyes and dry mucous membranes. Her serum calcium concentration is 16 mg/dL. Which of the following is the most appropriate next step in pharmacotherapy?
- ☐ A) Intramuscular calcitonin
 - ☐ B) Intravenous furosemide
 - ☐ C) Intravenous mithramycin
 - ☐ D) Intravenous pamidronate
 - ☒ E) Intravenous 0.9% saline

35. A 42-year-old woman with a 10-year history of rheumatoid arthritis comes for a follow-up examination. She has not taken any medications since discontinuing methotrexate 8 months ago after 2 years of continuous therapy. Medications used prior to that included nonsteroidal anti-inflammatory drugs, prednisone, and hydroxychloroquine. She reports that she feels better and does not need to take any medications. Examination shows subcutaneous nodules near the elbow. There is evidence of old joint destruction with swan-neck and boutonnière deformities; she has no active joint swelling or tenderness. The spleen tip is palpated 3 cm below the left costal margin. Laboratory studies show:

Hematocrit	36%
Leukocyte count	2000/mm ³
Segmented neutrophils	40%
Lymphocytes	60%
Platelet count	105,000/mm ³

Serum rheumatoid factor is present in high titers. Which of the following is the most likely cause of this patient's pancytopenia?

- ☐ A) Aplastic anemia
- ☐ B) Chédiak-Higashi syndrome
- ☐ C) Chronic lymphocytic leukemia
- ☐ D) Drug-induced myelofibrosis
- ☒ E) Felty syndrome

36. After 6 hours of chest pain, a 76-year-old man collapses at home. In the emergency department, a diagnosis of acute anterior wall myocardial infarction is made. His pulse is 103/min, and systolic blood pressure is 80 mm Hg by Doppler. He is ashen gray and has clammy skin. Diffuse bilateral pulmonary crackles are heard. Heart tones are muffled, and an S₃ is audible. Which of the following is the most likely cause of his hypotension?

- ☐ A) Third-degree atrioventricular block
- ☐ B) Decreased intravascular volume
- ☒ C) Decreased myocardial contractility
- ☐ D) Decreased ventricular filling pressure
- ☐ E) Pulmonary edema

37. A 42-year-old man comes to the physician because of a 1-month history of progressive shortness of breath. He has not had fever or cough productive of purulent or blood-tinged sputum. He does not smoke. He is receiving chronic oxygen for treatment of dyspnea. His younger brother had a single lung transplant for chronic obstructive pulmonary disease 1 year ago. The patient is 178 cm (5 ft 10 in) tall and weighs 63 kg (140 lb); BMI is 20 kg/m². His respirations are 24/min. Breath sounds are decreased. Bilateral basilar crackles are heard. An x-ray of the chest shows hyperlucent lung fields most prominent at the bases and flattened diaphragms. Which of the following is the most likely explanation for these findings?

- ☐ A) Abnormal alveolar macrophage chemotaxis
- ☐ B) ~~Abnormal chloride transport in apical membrane epithelium~~
- ☒ C) Abnormality of antiprotease resulting in destruction of alveolar supporting structures
- ☐ D) ~~Airway inflammation, obstruction, and hyperresponsiveness~~
- ☐ E) ~~Airway obstruction due to excessive, chronic, tracheobronchial mucus production~~
- ☐ F) Destructive changes in bronchial wall elastic and muscular layer

38. Twenty-four hours after admission to the hospital for treatment of community-acquired pneumonia, a 72-year-old woman's mental status has been fluctuating between barely arousable and alert and oriented. Current medications include azithromycin and ceftriaxone. At 4 am today, she reported to nursing staff that there were men in her room who were trying to steal her possessions and that she was afraid of animals outside her window who were trying to hurt her. Two hours later, she told the physician that she would like to go home. She said, "I know I have pneumonia and haven't been myself, but I'm feeling better now and would like to continue treatment as an outpatient." Vital signs are within normal limits. Crackles are heard at the right lung base. On mental status examination, she is pleasant, alert, and oriented to person, place, and time. Her Mini-Mental State Examination score is 25/30. She repeats that she really wants to go home and will sign out against medical advice (AMA) if the physician does not discharge her. Which of the following is the most appropriate next step?
- ☐ A) Allow the patient to sign out AMA and document that she should have remained in the hospital
 - ☐ B) Ask the patient's family to persuade her to remain in the hospital
 - ☒ C) Assess the patient's decision-making capacity
 - ☐ D) Contact the patient's family to determine if they are comfortable taking her home
 - ☐ E) Discharge the patient and schedule a follow-up examination in 2 days

39. Over the past 3 months, a 73-year-old man has had syncopal episodes that occur without warning. He has no history of illness. He takes no medications. Examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☒ A) Ambulatory ECG monitoring
- ☐ B) X-rays of the chest
- ☐ C) CT scan of the head
- ☐ D) Carotid Doppler scan
- ☐ E) Lumbar puncture

40. A 48-year-old man has had upper abdominal pain, nausea, and vomiting for 2 days. He has had a productive cough for 3 months. He has alcoholism and drinks a 12-pack of beer daily. He has smoked two packs of cigarettes daily for 30 years. His temperature is 38.4°C (101.1°F), pulse is 120/min, respirations are 16/min, and blood pressure is 90/55 mm Hg. Coarse breath sounds are heard on auscultation of the lungs. Cardiovascular examination shows no abnormalities. Abdominal examination shows tenderness with voluntary guarding in the left upper quadrant; bowel sounds are absent. Test of the stool for occult blood is positive. His serum calcium concentration is 7.6 mg/dL, and serum albumin concentration is 3.5 g/dL. Which of the following is the most likely diagnosis?

- | | |
|---|--|
| ☐ A) Addison disease | ☐ H) Metastatic malignancy |
| ☐ B) Hyperparathyroidism, primary | ☐ I) Milk-alkali syndrome |
| ☐ C) Hypoalbuminemia | ☒ J) Pancreatitis |
| ☐ D) Hypomagnesemia | ☐ K) Rhabdomyolysis |
| ☐ E) Hypoparathyroidism | ☐ L) Sarcoidosis |
| ☐ F) Lithium carbonate toxicity | ☐ M) Tuberculosis |
| ☐ G) Massive cell lysis | ☐ N) Vitamin D deficiency |

41. In diagnosing acute myocardial infarction, the upper limit of normal for serum creatine kinase (CK) MB fraction is increased from less than 5% total CK to less than 8% total CK. Which of the following is the effect on sensitivity and specificity?

- | | Sensitivity | Specificity |
|-------------------------------------|--------------------|--------------------|
| ☐ A) | Decreased | decreased |
| ☒ B) | Decreased | increased |
| ☐ C) | Increased | decreased |
| ☐ D) | Increased | no change |
| ☐ E) | No change | increased |

42. A 62-year-old man with hereditary hemochromatosis and cirrhosis is brought to the emergency department 30 minutes after his wife found him unconscious in bed. His cirrhosis has been complicated by ascites and hepatic encephalopathy. Current medications include spironolactone and lactulose. On arrival, he moans and withdraws all four extremities to painful stimuli but does not respond to commands. His temperature is 38.2°C (100.8°F), pulse is 130/min and regular, respirations are 24/min, and blood pressure is 70/40 mm Hg. Examination shows a supple neck. The lungs are clear to auscultation. Heart sounds are normal with no murmurs. Abdominal examination shows firmness and distention with shifting dullness; he moans on palpation of the abdomen. Bowel sounds are decreased. There is 2+ edema to the mid thigh. Test of the stool for occult blood is negative. Laboratory studies show:

Hematocrit	32%
Leukocyte count	17,500/mm ³
Segmented neutrophils	82%
Bands	10%
Lymphocytes	8%
Platelet count	125,000/mm ³
Prothrombin time	14.6 sec
	(INR=1.4)
Partial thromboplastin time	33 sec

Pulse oximetry on room air shows an oxygen saturation of 96%. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Echocardiography
- ☐ B) CT scan of the abdomen
- ☐ C) Esophagogastroduodenoscopy
- ☐ D) Liver biopsy
- ☐ E) Lumbar puncture
- ☒ F) Abdominal paracentesis

43. A 62-year-old man has the sudden onset of left hemiparesis that resolves completely in 2 hours. His blood pressure is 140/90 mm Hg. A right carotid bruit is heard. Examination, a complete blood count, an ECG, and a CT scan of the head show no other abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Ambulatory ECG monitoring
- ☐ B) Echocardiography
- ☐ C) Carotid arteriography
- ☒ D) Duplex ultrasonography of the carotid arteries
- ☐ E) MRI of the brain

44. A 20-year-old man comes to the emergency department 8 hours after the onset of severe pain, swelling, and discharge from an area over his lower back. He has a 1-month history of increasing pain in this area with sitting or activity, and the pain became exacerbated during a 400-mile car trip today. He has no history of serious illness and takes no medications. Examination shows an erythematous, fluctuant area of edema in the intergluteal cleft above the sacrum at the midline. The area is tender to palpation, and there is a small amount of purulent discharge from a small opening. Which of the following is the most appropriate immediate next step in management?

- ☐ A) Sitz baths
- ☐ B) Topical application of an antibiotic
- ☐ C) Examination under anesthesia
- ☒ D) Surgical drainage
- ☐ E) Wide excision

45. A 62-year-old woman comes to the physician because of urinary incontinence for 3 weeks. Sometimes she is unable to reach a bathroom quickly enough after feeling the need to void. She has hypertension treated with hydrochlorothiazide. She stopped taking the hydrochlorothiazide 6 days ago because she thought it was causing the incontinence, but the symptoms have continued. She takes no other medications. She underwent a hysterectomy for leiomyomata uteri 10 years ago. Her blood pressure is 140/92 mm Hg. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy for her incontinence?
- ☐ A) Bethanechol
 - ☐ B) Conjugated estrogens
 - ☐ C) Finasteride
 - ☒ D) Oxybutynin
 - ☐ E) Terazosin

46. A 47-year-old man comes to the physician because of a 1-month history of increasingly severe left hip pain. There is no history of trauma. He has been receiving hemodialysis for 6 years because of hypertensive nephropathy. He walks with a limp. X-rays of the left hip show a nondisplaced fracture of the femoral neck. Which of the following is the most likely underlying cause of these findings?
- ☐ A) Bacterial release of cytokines
 - ☐ B) Decreased peripheral vascular perfusion
 - ☒ C) Deficient mineralization of osteoid
 - ☐ D) Hematogenous bacterial seeding of the hip joint
 - ☐ E) Monocytic phagocytosis of crystals
 - ☐ F) Synovitis of the hip joint

47. A 62-year-old woman is brought to the physician by her husband because of a 2-day history of confusion. She has hypertension and type 2 diabetes mellitus. Current medications include ramipril and glipizide. She is oriented to person but not to place or time. Her temperature is 37°C (98.6°F). The remainder of the examination, including neurologic examination, shows no abnormalities. Laboratory studies show:

Hematocrit	24%
Leukocyte count	3400/mm ³
Segmented neutrophils	65%
Lymphocytes	35%
Serum	
Ca ²⁺	13 mg/dL
Creatinine	2 mg/dL
Protein, total	9.5 g/dL
Albumin	4.5 g/dL

Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the head
- ☐ B) Cefepime therapy
- ☒ C) Intravenous administration of 0.9% saline
- ☐ D) Lumbar puncture
- ☐ E) Hemodialysis

48. A previously healthy 32-year-old woman comes to the physician because of a 3-month history of low-grade fever, aching joint pain, especially in her hips and knees, and a 4.5-kg (10-lb) weight loss. Ibuprofen partially controls her joint pain. She has had generalized fatigue for 4 weeks. She also has noticed that she occasionally has a bumpy rash on her chest after being outside in the sun. She takes no other medications. There is no family history of similar symptoms. Menses occur at regular 27- to 30-day intervals with occasional heavy flow. She has smoked one pack of cigarettes daily for 15 years. She does not have a history of illicit drug use. She follows a vegetarian diet. She is married. She is 168 cm (5 ft 6 in) tall and weighs 54 kg (120 lb); BMI is 19 kg/m². Her temperature is 37.8°C (100°F), pulse is 76/min, respirations are 14/min, and blood pressure is 120/70 mm Hg. Examination shows no rash or oral lesions. There are several 1-cm cervical lymph nodes and 1-cm axillary lymph nodes bilaterally. Cardiopulmonary examination shows no abnormalities. The abdomen is soft and nontender. There is no splenomegaly. The wrists, hips, and knees are mildly tender bilaterally. There is no peripheral edema. Laboratory studies show:

Hematocrit	33%
Leukocyte count	1500/mm ³
Segmented neutrophils	55%
Bands	1%
Eosinophils	1%
Lymphocytes	40%
Monocytes	3%
Platelet count	155,000/mm ³
Erythrocyte sedimentation rate	88 mm/h
Serum	
Urea nitrogen	18 mg/dL
Creatinine	0.7 mg/dL
Rheumatoid factor	negative
Antinuclear antibody	positive at 1:320 in a speckled pattern
Smith antibody	positive

Which of the following is the most appropriate next step in management?

- ☐ A) Highly active antiretroviral therapy
- ☐ B) Multivitamin supplementation
- ☒ C) Prednisone therapy
- ☐ D) Bone marrow biopsy
- ☐ E) Lymph node biopsy
- ☐ F) Chemotherapy

49. A previously healthy 67-year-old woman is admitted to the hospital because of fever, hypotension, and respiratory distress. Her temperature is 38.4°C (101.1°F), pulse is 112/min, respirations are 28/min, and blood pressure is 98/50 mm Hg. Diffuse crackles are heard on auscultation. A grade 2/6 systolic murmur is heard. The extremities are warm and well perfused. Pulmonary artery catheterization shows a cardiac output of 6 L/min (N=4–5), a left ventricular end-diastolic pressure of 11 mm Hg (N=5–12), and a pulmonary capillary wedge pressure of 14 mm Hg (N=5–16). Laboratory studies show:

Hematocrit	35%
Leukocyte count	18,000/mm ³
Segmented neutrophils	72%
Bands	8%
Lymphocytes	20%

Arterial blood gas analysis on room air:

pH	7.3
Pco ₂	18 mm Hg
Po ₂	64 mm Hg

Which of the following is the most likely mechanism of this patient's respiratory failure?

- ☐ A) Decreased hypoxic drive
- ☐ B) Decreased myocardial contractility
- ☐ C) Decreased pulmonary vascular flow
- ☐ D) Increased pericardial pressure
- ☐ E) Increased regurgitant mitral flow
- ☒ F) Increased vascular permeability



50. For 2 months, a 27-year-old woman has had hip pain that is most severe with weight bearing. She takes ibuprofen occasionally for headaches and methylprednisolone for systemic lupus erythematosus. There is no other history of serious illness. She walks with a marked limp. Passive and active movement of the hip causes pain. An x-ray is shown. Which of the following is the most likely diagnosis?

- ☒ A) Osteonecrosis
- ☐ B) Reactive arthritis
- ☐ C) Rheumatoid arthritis
- ☐ D) Septic arthritis
- ☐ E) Stress fracture
- ☐ F) Transient osteoporosis